

On the Invariant Theory of Forms in n Variables

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§ 5. The Decomposition Identities in Explicit Form.

If, in case 4) of (25), we specialize τ to $\sigma + \rho$, we obtain

$$(q_\rho A^\sigma \mid y^{(1)} \dots y^{(\rho+\sigma)}) = \sum_{i_\rho=i_{\rho-1}+1}^{\tau} \varepsilon_{i_\rho}(q_\rho \mid y^{(i_1)} \dots y^{(i_\rho)})(A^\sigma \mid y^{(i_{\rho+1})} \dots y^{(i_{\rho+\sigma})}), \quad (32)$$

where $\varepsilon_{i_\rho} = (-1)^{\delta_{i_\rho}}$ and $\delta_{i_\rho} = \rho(\rho+1)/2 + i_1 + \dots + i_\rho$. This is a more general form of the product theorem for matrices than (16) and (17), and follows by Laplace expansion of the determinant of the product matrix

$$\sum \pm (v^{(1)} y^{(1)}) \dots (v^{(\rho)} y^{(\rho)})(a^{(1)} y^{(\rho+1)}) \dots (a^{(\sigma)} y^{(\rho+\sigma)}).$$

Thus the more general product theorem is composed from identities (20) and (21), respectively, and this explains the extraction of the special forms (16) and (17).

Relation (32) now provides the means for representing the decomposition identities explicitly. To this end we regard the forms occurring in (31) as ground forms; that is, we perform substitution (11), which changes nothing in the explicit representation in the rows A, B , since, by (8), the newly introduced rows R can be expressed in terms of the rows A, B themselves and not in terms of their component rows $a^{(1)} a^{(2)} \dots, b^{(1)} b^{(2)} \dots$. Thus let

$$(q_{\rho-\alpha} B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma} p_{\tau-\alpha}) = (R_{\rho-\alpha} p_{n-\rho+\alpha})(R^{\tau-\alpha} \mid p_{\tau-\alpha}). \quad (33)$$

Then the individual sum of (25) corresponding to the index α becomes

$$\begin{aligned} & \sum \varepsilon_{i_\alpha} (R_{\rho-\alpha} p_{n-\rho} y^{(i_1)} \dots y^{(i_\alpha)})(R^{\tau-\alpha} \mid y^{(i_{\alpha+1})} \dots y^{(i_\tau)}) \\ &= \sum \varepsilon_{i_\alpha} ([R_{\rho-\alpha} p_{n-\rho}]^\alpha y^{(i_1)} \dots y^{(i_\alpha)})(R^{\tau-\alpha} \mid y^{(i_{\alpha+1})} \dots y^{(i_\tau)}) \\ &= \varepsilon([R_{\rho-\alpha} p_{n-\rho}]^\alpha R^{\tau-\alpha} \mid p_\tau) = \varepsilon(R^{\tau-\alpha} q_{n-\tau} \mid R_{\rho-\alpha} p_{n-\rho}). \end{aligned} \quad (34)$$

An explicit final expression has therefore indeed been obtained, and the symmetric form in which q_ρ and p_τ occur shows that (30) leads to the same final expression, which is consequently independent of the original decomposition. A difference between (25) and (30) exists only as long as the rows y and v have independent significance, so that the decomposition identity passes, according to our definition, into a decomposable identity. To obtain identities with more symbol rows, one need only, by § 3 (end), resolve a row q_ρ into $q_{\rho_1} C^{\sigma_1}$, or p_τ into $p_{\tau_1} C_{\tau-\tau_1}$. The resulting expressions

$$(q_{\rho_1} C^{\sigma_1} \mid [R^{\tau-\alpha} q_{n-\tau}]_\lambda R_{\rho_1+\sigma_1-\alpha-\lambda}), \quad ([R_{\rho-\alpha} p_{n-\rho}]^\lambda R^{\tau-\alpha} \mid p_{\tau_1} C_{\tau-\tau_1}) \quad (35)$$

are exactly of the type occurring with two symbol rows, and therefore again admit explicit representation after a substitution corresponding to (33). Repetition consequently yields an explicit representation for arbitrarily many rows. It may also be noted that the first and last sums in (25) and (30), respectively, yield the explicit representation without applying (33), solely by formula (32), or else reduce altogether to a single term containing all rows explicitly. The relation obtained from (32) by resolving q_ρ into $q_{\rho_1} C^{\sigma_1}$, and so forth, may be regarded as the product theorem in its most general form.¹

In summary, we shall show:

Theorem III. With the identities

$$(q_\rho A^\sigma \mid p_\tau B_{\rho+\sigma-\tau}) = \sum_{0, \tau-\sigma}^{\tau, \rho} \varepsilon(R^{\tau-\alpha} q_{n-\tau} \mid R_{\rho-\alpha} p_{n-\rho}), \quad (36)$$

¹In the original print, the two expressions in (35) use the component index α , with the first containing $[R^{\tau-\alpha} q_{n-\tau}]_\alpha R_{\rho_1+\sigma_1-\alpha}$. The current R823 German source makes the coordinated editorial change from α to λ in both expressions and correspondingly uses $[R^{\tau-\alpha} q_{n-\tau}]_\lambda R_{\rho_1+\sigma_1-\alpha-\lambda}$ in the first; R823 is followed here.

where the rows R are determined by (33), together with those obtained from them by resolving q_ρ, p_τ into $q_{\rho_1} C^{\sigma_1}$, and so forth, the totality of indecomposable identities is exhausted. In this combined-bound notation, as in the preceding section, the summation index runs from the larger lower value to the smaller upper value.

Indeed, the resulting identities are the most general of their kind, since the individual rows are no longer subject to any restriction with respect to weight or number, as was still the case in (20) and (21). Nor do any further identities among these rows exist. For, on resolving a row p_τ into $y^{(1)} \dots y^{(\tau)}$, the most general identities are composed from (20), and hence, by § 3 (end), from (20a); the composition discussed after (16) is precisely the one carried out in § 4. The passage to arbitrarily many rows is supplied, as the discussion of (20a) shows, by applying the same operation repeatedly to the expressions arising from the resolution of q_ρ into $q_{\rho_1} C^{\sigma_1}$, and so forth. It follows as well that the direct passage from (20), instead of from (20a), yields nothing new. This is easily verified computationally: specialize a row q_ρ in (25) once, and another time carry out the passage from (20) by the method of § 4. The same sums arise both times, and application of the general product theorem (see above) transforms them into the identities obtained from (36). The identities (20) and (21) correspond to the special cases $\tau = 1$ and $\rho = 1$; only two individual sums then occur in each case, hence, in accordance with the preceding remark, an explicit representation without substitution (33), in agreement with the earlier result.

Finally, let us briefly mention two special cases of the decomposition identity. In (31), set $\tau = \sigma$, $\rho = n - \sigma$, and denote the row p_τ by $p'_\sigma \sim q'_{n-\sigma}$. Then

$$(A^\sigma | p_\sigma)(B^\sigma | p'_\sigma) - (A^\sigma | p'_\sigma)(B^\sigma | p_\sigma) = \sum_{\alpha=1}^{\sigma, n-\sigma} \Pi(q_{n-\sigma-\alpha} B^\sigma | A_{n-\sigma} p_{\sigma-\alpha}). \quad (37)$$

This is Clebsch's formula² for expanding a form with two cogredient rows p_σ, p'_σ in elementary covariants. The elementary covariants belonging to a form $(A^\sigma | p_\sigma)^m (B^\sigma | p'_\sigma)^n$ are then the following:

$$\left\{ \sum_{\alpha=1}^{\sigma, n-\sigma} (q_{n-\sigma-\alpha} B^\sigma | A_{n-\sigma} p_{\sigma-\alpha}) \right\}^\rho (A^\sigma | p_\sigma)^{m-\rho} (B^\sigma | p'_\sigma)^{n-\rho}, \quad (\rho = 0, 1, \dots, m, n). \quad (38)$$

Here the paired upper endpoints again mean the smaller of the two displayed values; likewise, the final range for ρ ends at the smaller of m and n . These covariants arise, as we shall briefly say, from the ground form by “cogredient folding of defect ρ ” (cf. § 2).³

Second, set $\tau = \sigma - \lambda$, $\rho = n - \sigma - \lambda$, and $B^\sigma = A^\sigma$. Then

$$\begin{aligned} (q_{n-\sigma-\lambda} A^\sigma | A_{n-\sigma} p_{\sigma-\lambda}) &= (-1)^\lambda (q_{n-\sigma-\lambda} A^\sigma | A_{n-\sigma} p_{\sigma-\lambda}) \\ &+ (-1)^{(n-\sigma)(\sigma-\lambda)} \sum_{\alpha=1}^{\sigma-\lambda, n-\sigma-\lambda} \Pi(q_{n-\sigma-\lambda-\alpha} A^\sigma | A_{n-\sigma} p_{\sigma-\lambda}). \end{aligned} \quad (39)$$

Thus, as noted in § 2, for odd defect λ the conditions (12) characterizing the symbol rows are consequences of the conditions of higher defect. For a linear form $(A^\sigma | p_\sigma) = (B^\sigma | p_\sigma)$,⁴ it follows from (39) that its irreducible invariant formations that are quadratic in the coefficients reduce to ones of even defect. This agrees with the well-known fact that a linear form in the binary and ternary domains possesses no invariant formations.

It should be noted that the general identities were originally derived under the assumption that the individual rows satisfy conditions (12). Since, however, these individual rows enter the identities only linearly, the latter hold also for the more general rows that do not satisfy conditions (12).

²Clebsch, loc. cit., pp. 25–32. Clebsch shows that the individual terms on the right-hand sides depend only on the rows A, B ; the explicit representation would result by applying (32) to his expressions Φ_h .

³The original print reads “defect α ” in this sentence. The current R823 German source reads “defect ρ ”; R823 is followed here because ρ is the outer exponent and the immediately stated range parameter in (38). This is a disclosed print/R823 editorial delta, not a claim that the printed reading is mathematically erroneous.

⁴The original print places the identity sign $\equiv$ between these two expressions; the current R823 German source uses ordinary equality. R823 is followed here, with the historical notation preserved in this disclosure. This is recorded as an editorial notation delta, not as a proved mathematical source defect.